

and the solar industry too is passing through that stage. Some of these are:

**Lack of skilled manpower.** A major problem that India is likely to face in the future is, the shortage of skilled manpower. Sudeep Jain, chairman and managing director, Tamil Nadu Energy Development Agency (TEDA) says, "Finding good manpower for a few thousand installations every week is a major challenge, especially when TEDA is gearing up to install 300,00 home and 100,00 street solar lighting systems in the next five years." Experts think that solar-specific training in aspects like technical, detailed project reports, feasibility studies, design, technology selection and operation and maintenance is a requisite, although a B.Tech program in solar seems to be far fetched.

TEDA has planned to put up separate training sessions for the aspiring professionals in solar at its annual renewable energy event, Renergy, in Chennai. While GSES, a training organisation, has developed courses to cover various aspects of PV installation, the courses also cover sales, installation and maintenance of PV systems. One of their training course is 'Standalone Solar Power Supply Systems: Design and Installation' that is now offered in India. It covers all aspects of off-grid PV installation, including key issues in site assessment, quality and performance issues of solar PV modules, shadow analysis and planning and solar radiation.

**Less number of retail products.** Manufacturers state that although solar products like emergency lights, PV modules, lanterns and water heaters are available both in urban centres and villages at retail shops, their numbers are quite less. For instance, in Delhi, such products are available only at shops in the electrical market of Chandni Chowk. Solar shops like Akshay Urja (previously known as Aditya Solar Shops) that opened some time ago are either not doing good business or have shut down. Tarun Kapoor, joint secretary, MNRE, at a seminar recently said, "India's lack of retail outlets for solar

products is hampering sales to a large extent. We need to develop a mechanism wherein solar products are available in the open market so that consumers can buy these products easily."

"Since solar products do not get categorised as part of consumer electronics, we cannot expect them to be available at the typical consumer electronics shops. Although there are retail shops in cities and villages, their number is quite meagre. But with the demand rising, the number of such shops will grow," says Dr Satyendra Kumar, chairman, Surya EnerTech. He adds, "Manufacturers are delivering their products to the market through their distributors, who are taking them to the retail shops. Retailers also approach the manufacturers directly for such products."

Until now it was the government alone that was boosting the growth of solar products, but now the scenario is changing. Seeing the demand for solar products grow in rural areas and in places with rampant power outage, manufacturers are either opening their own retail shops or tying with other retail outlets. Many other actions are also being taken care of, like Essmart Global has converted many of its kirana shops into retail shops for solar products. Anu Solar Pvt Ltd, Bengaluru-based solar product manufacturers and traders, have opened many retail shops for solar products. The store called Anu World has products like solar-powered calculators, caps with fans that run on solar

**TABLE II**  
**Solar Renewable Purchase Obligation by JNNSM**

| Year    | Energy demand (MU)* | Solar RPO (%) | Solar energy requirement (MU) for RPO compliance | Solar capacity requirement for RPO compliance (MW) |
|---------|---------------------|---------------|--------------------------------------------------|----------------------------------------------------|
|         | (A)                 | (B)           | (A)X(B)                                          |                                                    |
| 2011-12 | 953,919             | 0.25          | 2385                                             | 1433                                               |
| 2012-13 | 1,022,287           | 0.25          | 2556                                             | 1536                                               |
| 2013-14 | 1,095,555           | 0.50          | 5478                                             | 3291                                               |
| 2014-15 | 1,174,074           | 0.75          | 8806                                             | 5291                                               |
| 2015-16 | 1,258,221           | 1.00          | 12,582                                           | 7560                                               |
| 2016-17 | 1,348,399           | 1.25          | 16,855                                           | 10,127                                             |
| 2017-18 | 1,443,326           | 1.75          | 25,258                                           | 15,176                                             |
| 2018-19 | 1,544,936           | 2.25          | 34,761                                           | 20,885                                             |
| 2019-20 | 1,653,700           | 2.50          | 41,343                                           | 24,839                                             |
| 2020-21 | 1,770,120           | 2.75          | 48,678                                           | 29,247                                             |
| 2021-22 | 1,894,736           | 3.00          | 56,842                                           | 34,152                                             |

Source: Novonious Insights

**TABLE III**  
**Solar PV Cost-Reduction Drivers**

| Module-Level                   | 2010  | 2012   | 2016  |
|--------------------------------|-------|--------|-------|
| Wafer thickness (Micro meters) | 200   | 180    | 125   |
| Silver per cell (grams/cell)   | 0.3   | 0.23   | 0.1   |
| Cell efficiency                | 14%   | 16.50% | 20%   |
| Glass absorption               | 4%    | 3%     | 1.75% |
| Glass reflection               | 0.98% | 0.80%  | 0.55% |
| Glass thickness (mm)           | 3.2   | 3.2    | 2.85  |
| Module to cell power ratio     | 97%   | 98.50% | 100%  |

Source: International Technology Roadmap for Photovoltaic (ITRPV) and other estimates

power, solar water heaters and solar power based storage batteries and inverters. The significant matter to note is that, Anu World is a channel partner to MNRE and sells its products at subsidised rates.

Solar service providers, named Solar Basket, are all set to open roughly 150 stores across India by the end of this year. They claim to have about eight stores and 20 franchisees pan-India currently. So soon the cities like Guwahati, Vijayawada, Hyderabad, Itanagar, Chandigarh, Bilsapur, Surat, Panaji and Gurgaon will have solar stores too.

**Not enough capital.** Another im-